UG (NEP) 7th SEMESTER
APPLIED COMPUTING: DESIGN & ANALYSIS OF ALGORITHMS
(ACP722N)

No. of Questions: 09 Session: Month–Year (not specified on paper)
Time Allowed: 2:00 Hours Max. Marks: 75 Min. Marks: 27

NOTE: Attempt all questions from Section “A” and “B” and only two questions from Section “C”.

Section A: Short Answer Type Questions (8 × 1.5 = 12 Marks)

Answer all questions in a maximum of 20 words each.

i. Define Space Complexity.
ii. What is Big-O notation?
iii. Define Knapsack Problem.
iv. Define Insertion Sort.
v. What is time complexity of linear search?
vi. State worst-case time complexity of Bubble Sort.
vii. What is Backtracking?
viii. Define Branch and Bound strategy.

Section B: Medium Answer Type Questions (4 × 7.5 = 30 Marks)

Answer all the four questions with internal choice in a maximum of 150 words each.

2. Explain Growth of Functions with examples. Why is it important in algorithm comparison?

OR

Write the recursive algorithm for Tower of Hanoi and mention its time complexity.

3. Explain Binary Search with an example. Mention best, worst, and average time complexity.

OR

Explain Quick Sort with pivot selection and complexity.

4. Discuss optimal substructure property in Dynamic Programming.

OR

Write short note on 8-Queens problem. How is backtracking applied to solve it?

5. Explain Dynamic Programming and its main characteristics.

OR

Discuss optimal substructure property in Dynamic Programming.

Section C: Long Answer Type Questions (2 × 16.5 = 33 Marks)

Answer only two questions in a maximum of 400 words each.

6. Define Big-O, Big-Ω, Big-Θ. Discuss their significance in algorithm analysis with graphical representation and examples.
7. Explain various searching algorithms. Compare their performance.
8. Discuss P, NP, NP-complete, NP-hard classes. Explain polynomial-time reductions and their significance in complexity theory.
9. State and explain Master’s Theorem. Use it to solve simple recurrence relations with examples.